

# The two channels of sequence dependence

## the prelude and what expands it

### 1. The prelude

For the introduction, immediately after the opening sentence saying that how sequence dependence registers in observed statistics has received little attention, and before the paper’s own contribution is stated (plan 1.A2). Approved 2026-09-27, with the settings made explicit the same day.

More specifically, arrival-sequence dependence can enter through two channels, an **association** channel, in which one cue changes what the other implies and which exists only when the attributes are believed related, and a **position** channel, in which the observer weights the later cue less whatever the attributes are (HogarthEinhorn1992,Asch1946). Which channels operate is fixed by how fully the later cue is adopted. When it is adopted only in part, the position channel operates and every marginal registers the sequence even when the attributes are believed unrelated. When each impression is adopted in full (DiaconisZabell1982), the position channel is absent, the association channel is the only one, and the sequence registers unevenly across statistics. The current paper explores the full-adoption setting. Table ?? in Section 6 sets the two settings side by side.

The prelude does four things, and each section below expands one of them. It names the two channels (Sections 4 and 5 below give their mechanics). It says that the adoption weight decides which channels operate (Section 3). It says what each setting does to the statistics (the table it cites, Section 2). And it says which setting the paper studies (Section 6, the two paragraphs for the scope section).

### 2. The two settings, the table the prelude cites

For Section 6 of the manuscript, beside the two paragraphs of plan 6.B, labelled **tab:settings**. Every cell is a proposition already in the manuscript or a row of the independence table (Section 4 below). The rows of the full comparison (Section 5) that are placements onto Weisberg’s vocabulary, and the decision row, which is inferred, are not in it. The distinction in the first row, order dependence arising in the rule or in the inputs, is Weisberg’s; his name is kept out of the cells because the preprint carries no bibliographic line, and can be added to the caption once the entry is confirmed.

Table 1: The two settings. Under full adoption only the association channel operates; under partial or no adoption the position channel operates as well, and the association channel with it when the attributes are believed related.

|                                   | <b>Full adoption, <math>\omega = 1</math></b><br>(association channel only)                                                                                      | <b>Partial or no adoption, <math>\omega &lt; 1</math></b><br>(position channel operates)                    |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| Where the order dependence arises | In the rule. The inputs $q, r$ are the same in either position; order matters because two rigid steps on correlated partitions do not commute (Diaconis–Zabell). | In the inputs. What the second cue delivers depends on its position; the rule adds nothing new at $c = 0$ . |
| Source in the model               | Association only. Zero at $c = 0$ (Proposition IMM).                                                                                                             | Position, present at $c = 0$ , plus association when $c \neq 0$ .                                           |

|                                | Full adoption, $\omega = 1$                                                             | Partial or no adoption, $\omega < 1$                                                                                                                                                                                                                                                     |
|--------------------------------|-----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Status of the inputs           | Order-independent. The credential delivers $q$ whether or not the letter was read.      | Order-dependent. The later cue is adopted only in part, so what it delivers depends on being second.                                                                                                                                                                                     |
| Size of the effect             | Small, first order in $c$ , spread unevenly across statistics.                          | Blunt. $(1 - \omega)(\alpha - q_0)$ on a marginal whatever $c$ is (Proposition ADJ).                                                                                                                                                                                                     |
| Marginals                      | Last-read exact, first-read drifts at first order (Proposition DRF).                    | Both differ at zeroth order between sequences.                                                                                                                                                                                                                                           |
| Believed association           | Second order, between sequences and pooled (Propositions DEC, PRO).                     | Zero between sequences at $c = 0$ , first order in $c$ . Pooled, nonzero at $c = 0$ , $-\lambda(1 - \lambda)(1 - \omega)^2(\alpha - q_0)(\beta - r_0)$ .                                                                                                                                 |
| Odds ratio                     | Identical across sequences, every $c$ (Lemma SEP).                                      | Identical across sequences, every $c$ (Lemma SEP). The one statistic that does not move.                                                                                                                                                                                                 |
| Classification by order in $c$ | Is the contribution. Which statistics are protected, and why.                           | Shifts by one order with the ranking unchanged. Marginals at zeroth order, protected statistics at first order, the odds-ratio shadow at second, the odds ratio never (Proposition LAD; corrected 2026-10-01, the earlier “is moot” is refuted by <code>sympy/verify_ladder.py</code> ). |
| Audit implication              | False negative. An association audit passes a sequence-dependent population.            | False positive as well. A pooled association audit returns a stereotype no member holds, from mixing alone.                                                                                                                                                                              |
| Identification                 | The reading, from which marginal ignores the prior association (Propositions ORD, DRF). | The weight, from three ratings (Proposition ADJ). At its strongest here, since the position channel is what ADJ reads.                                                                                                                                                                   |

### 3. Where each clause of the prelude gets its numbers

The prelude’s clauses, in order, and the proposition or paragraph that expands each.

Table 2: The prelude clause by clause.

| Prelude clause                                                                   | Quantitative form                                                                                                                                                     | Where it lives                           | Still to add                                                                                                                                                                                      |
|----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| an association channel that exists only when the attributes are believed related | $P_{AB}^J - P_{BA}^J = c(\kappa R_1 - \kappa' R_2) + \mathcal{O}(c^2)$ ; zero at $c = 0$ ; zero only when each delivered credence equals the prior marginal it resets | DIV and IMM, Section 4 of the manuscript | Optionally the forcing sentence: with impressions read as delivered credences, the effect is nonzero at first order whenever either cue moves its marginal (at 3.B’s spot, or a remark after DIV) |
| a position channel that weights the later cue less                               | the three lines of ADJ; $\omega$ from three ratings                                                                                                                   | 5.B, Proposition ADJ                     | Nothing                                                                                                                                                                                           |

| Prelude clause                                                                        | Quantitative form                                                                                                                                                                                                                                                       | Where it lives                                                                                | Still to add                                                                                              |
|---------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| which channels operate is fixed by how fully the later cue is adopted                 | last-read marginal pinned at $\omega = 1$ (ADJ endpoint); the association's sequence-effect $c^1$ coefficient carries $(1 - \omega)$ ; the pooled association distortion carries $(1 - \omega)^2$ ; a position-scaled Bayes-factor update commutes only at $\omega = 1$ | rows 3–5 of the interior script; Table ??; the two-channel paragraph (Section 6 below)        | The two-channel paragraph. It is the only place this clause is expanded in prose.                         |
| every marginal registers the sequence even when the attributes are believed unrelated | $(1 - \omega)(\alpha - q_0)$ on competence, $-(1 - \omega)(\beta - r_0)$ on trustworthiness, at $c = 0$                                                                                                                                                                 | ADJ gives the first; the second is row 1 of Table ?? and appears in the two-channel paragraph | Nothing. Decided 2026-09-27: the trustworthiness value lives in the Section 6 paragraph, not in ADJ.      |
| registers unevenly across statistics                                                  | ORD's row table, $c\kappa'$ , $-c\kappa$ , $\mathcal{O}(c^2)$ on the association; PRO's criterion; the manuscript's summary table                                                                                                                                       | 5.B and the summary table, Section 5 of the manuscript                                        | Nothing                                                                                                   |
| the current paper explores the full-adoption setting                                  | the scope sentences, the two channels named, the rubric prediction                                                                                                                                                                                                      | Section 6 of the manuscript, the two paragraphs of 6.B, plus the Assumption 3 scope sentence  | The Assumption 3 sentence, if the one-cue-per-attribute principle is to be stated where the assumption is |
| Table ?? sets the two settings side by side                                           | the ten rows of Table ??                                                                                                                                                                                                                                                | Section 6 of the manuscript, beside 6.B                                                       | The table itself (Section 2 above)                                                                        |

## 4. The position channel at independence, the verified basis

This section is behind the right-hand column of Table ?? and behind the prelude's clause that under partial adoption every marginal registers the sequence even when the attributes are believed unrelated. Setting: the prior covariance is  $c = 0$ , so the two attributes are believed unrelated, and  $\omega < 1$ , so the cue read second is adopted in part or not at all. At  $c = 0$  the association channel is switched off; what remains is the position channel at full strength. All quantities are exact at  $c = 0$ ; the  $\mathcal{O}(c)$  corrections are in the script and are not the point.

Table 3: What can be said at  $c = 0$  and  $\omega < 1$ .

| Statistic                                       | Value at $c = 0$                                                                                                             | What it says                                                                                                                                                                                                                                                                                                                      |
|-------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Between the two sequences, one evaluator</i> |                                                                                                                              |                                                                                                                                                                                                                                                                                                                                   |
| 1 marginal probabilities                        | $P_{AB}^J(A=1) - P_{BA}^J(A=1) = (1 - \omega)(\alpha - q_0)$<br>$P_{AB}^J(B=1) - P_{BA}^J(B=1) = -(1 - \omega)(\beta - r_0)$ | The marginals carry the whole effect. Each is the distance the cue would have moved its marginal, scaled by how much of it was not adopted. Zero only at $\omega = 1$ or when the cue delivers the prior marginal. The mechanism is position, not association: in $AB$ the letter is the weakened cue, in $BA$ the credential is. |

|                                                                                            | Statistic                                                     | Value at $c = 0$                                                                                                 | What it says                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|--------------------------------------------------------------------------------------------|---------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2                                                                                          | believed association (cross-product; odds ratio, Yule's $Q$ ) | $\text{assoc}(P_{AB}^J) - \text{assoc}(P_{BA}^J) = 0$<br>odds ratio identical, for every $c$                     | Both sequences leave a product table at $c = 0$ , so the cross-product is zero in each; the odds ratio equals the prior's because every damped step is a row or column rescaling (Lemma SEP). The paper's qualitative verdict survives: the marginals see the sequence, the association does not. What changes is the order, zeroth instead of first, and with it the relevance of small $c$ .                                                                                                                                                                                                                                        |
| 3                                                                                          | the adoption weight                                           | $\omega = 1 - \frac{P_{AB}^J(A=1) - P_{BA}^J(A=1)}{\alpha - q_0}$                                                | Row 1 read backwards. The channel the paper does not study is the channel Proposition ADJ reads; the identification device lives entirely at this point. Requires $q_0 \neq \alpha$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 4                                                                                          | decision flip and loss                                        | score gap = $\langle v, P_{AB}^J - P_{BA}^J \rangle$ , a fixed multiple of row 1, independent of $c$             | The share of evaluators whose decision flips is the mass of a band of fixed width, not $\mathcal{O}(c)$ , and the surplus-weighted loss is not second order. <i>Inferred from linearity of the score; not separately verified. Not in Table ??.</i>                                                                                                                                                                                                                                                                                                                                                                                   |
| <i>Pooled over a population, against the benchmark, <math>0 &lt; \lambda &lt; 1</math></i> |                                                               |                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 5                                                                                          | association of the mean belief                                | $\text{assoc}(\bar{P}_\lambda) - \text{assoc}(P^B) = -\lambda(1-\lambda)(1-\omega)^2(\alpha - q_0)(\beta - r_0)$ | The mean belief acquires an association nobody holds. Every evaluator's table is independent and so is the benchmark's, but the mixture of two product tables with different marginals is not a product table. A pooled audit of whether the traits go together returns a nonzero value in a population where no one believes they do. The sign is fixed by whether the two cues move their marginals in the same direction relative to the prior. No counterpart at $\omega = 1$ , where the mixture at $c = 0$ is $q \otimes r$ .                                                                                                   |
| 6                                                                                          | marginals of the mean belief                                  | $\bar{P}_\lambda(A=1) - P^B(A=1) = (1-\lambda)(1-\omega)(q_0 - \alpha)$                                          | Zeroth order. $P^B$ sets both marginals to their delivered credences, so it is a full-adoption object under the other reading, and the gap from it mixes two differences, the reading and the adoption weight. No sequence-free reference exists at interior $\omega$ under either reading: a Bayes-factor update that scales the second cue's factor by $\omega$ is itself order-dependent at $c = 0$ (script, row 5), because Wagner's commutation needs the same factor in either position. The between-sequence comparison of rows 1–2 needs no reference, which is one more reason to prefer Proposition ORD's form of the test. |

## 5. The full comparison, with Weisberg’s placements (record)

This is Table ?? before pruning, kept as the record of what was set aside and why. The primacy regime is taken for the interior, since with  $\omega$  on the second cue every interior value weakens the later impression. Weisberg (*Commutativity or Holism?*, read in full 2026-09-27; see [literature/weisberg2009](#)) supplies the vocabulary. He separates the *rule* that propagates a change on a partition from the *inputs problem* of how an experience fixes the values on that partition; Lange’s point is that reversing two experiences does not reverse their input values; Wagner’s theorem, read as a dilemma, says that commutativity on experiences forces each experience to carry the same Bayes factor whatever background it meets, which is anti-holism; and rigidity, the holding of conditionals fixed, prevents a later discovery from ever *undercutting* earlier support, leaving only rebutting. Marks: † a row of Table ?? or of `verify_interior_omega.py`, in Table ??; ‡ inferred, not separately checked, not in Table ??; ◊ a mapping onto Weisberg’s terms, not a computation, not in Table ?? except for the first and third rows, which enter without his name.

Table 4: Full adoption against partial or no adoption, all rows.

|                                                  | Full adoption, $\omega = 1$                                                                                                                                                                                                                                                                                                                         | Partial or no adoption, $\omega < 1$                                                                                                                                                                                                                          |
|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Where the order dependence arises ◊              | In the rule (the association channel). The inputs $q, r$ are the same in either position; order matters because two rigid steps on correlated partitions do not commute (Diaconis–Zabell).                                                                                                                                                          | In the inputs (the position channel). The value the second cue delivers depends on its position; the rule adds nothing new at $c = 0$ .                                                                                                                       |
| Source in the model †                            | Association only. Zero at $c = 0$ (IMM).                                                                                                                                                                                                                                                                                                            | Position, present at $c = 0$ , plus association.                                                                                                                                                                                                              |
| Status of the inputs ◊                           | Order-independent. Anti-holistic at the inputs, in Weisberg’s sense: the credential delivers $q$ whether or not the letter was read.                                                                                                                                                                                                                | Order-dependent. History-dependent inputs, not holism in Weisberg’s sense: the import of the second cue depends on its position, not on the credence in any defeater proposition.                                                                             |
| Weisberg’s dilemma, Bayes-factor reading ◊       | The commutative horn. Field’s $\alpha$ , Wagner’s theorem, the benchmark $P^B$ .                                                                                                                                                                                                                                                                    | Off the commutative horn, but not on the holistic one. A position-dependent factor is order-dependent at $c = 0$ (row 5 of the script) because commutation needs the same factor in either position; the dependence is on position, not on a defeater belief. |
| Weisberg’s dilemma, delivered-credence reading ◊ | On neither horn. Inputs fixed regardless of any background, more anti-holistic than Field; non-commutative because the implied factor $r_j/P^A(B=j)$ varies with what the first cue did. This is non-commutativity on input distributions, the case Lange declines and Weisberg sets aside (p. 9). The two readings differ as much as they ever do. | Non-commutative through the inputs, as under the other reading; the readings differ only in how the surviving part of the earlier impression is blended, linearly in probability or geometrically in odds.                                                    |
| Rigidity ◊                                       | Both settings are rigid. Rigidity preserves independence, which is IMM at $c = 0$ for every $\omega$ ; a later cue can never undercut what an earlier one implied, only rebut it, and the only rebutting route is the prior association $c$ .                                                                                                       |                                                                                                                                                                                                                                                               |
| Size of the effect †                             | Small, first order in $c$ , spread unevenly across statistics.                                                                                                                                                                                                                                                                                      | Blunt. $(1 - \omega)(\alpha - q_0)$ on a marginal whatever $c$ is.                                                                                                                                                                                            |
| Marginals †                                      | Last-read exact, first-read drifts at first order (DRF).                                                                                                                                                                                                                                                                                            | Both differ at zeroth order between sequences.                                                                                                                                                                                                                |

|                                     | Full adoption, $\omega = 1$                                                         | Partial or no adoption, $\omega < 1$                                                                                                                                                                                                                                                     |
|-------------------------------------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Believed association<br>†           | Second order, between sequences and pooled (DEC, PRO).                              | Zero between sequences at $c = 0$ , first order in $c$ . Pooled, nonzero at $c = 0$ , $-\lambda(1 - \lambda)(1 - \omega)^2(\alpha - q_0)(\beta - r_0)$ .                                                                                                                                 |
| Odds ratio †                        | Identical across sequences, every $c$ .                                             | Identical across sequences, every $c$ . The one statistic that does not move.                                                                                                                                                                                                            |
| Decisions ‡                         | Share of flips first order, loss second order (SHR, LOS).                           | Score gap fixed in $c$ , so the share is a fixed band and the loss is not second order.                                                                                                                                                                                                  |
| Classification by<br>order in $c$ † | Is the contribution. Which statistics are protected, and why.                       | Shifts by one order with the ranking unchanged. Marginals at zeroth order, protected statistics at first order, the odds-ratio shadow at second, the odds ratio never (Proposition LAD; corrected 2026-10-01, the earlier “is moot” is refuted by <code>sympy/verify_ladder.py</code> ). |
| Audit implication †                 | False negative. An association audit passes a sequence-dependent population.        | False positive as well. A pooled association audit returns a stereotype no member holds, from mixing alone.                                                                                                                                                                              |
| Identification                      | The reading, from which marginal ignores the prior association (ORD, DRF).          | The weight, from three ratings (ADJ). At its strongest here, since the position channel is what ADJ reads.                                                                                                                                                                               |
| Literature stance ◊                 | Diaconis–Zabell’s setting; Hawthorne’s objection inherited; Lange’s point declined. | Hawthorne answered by measurement; Hogarth–Einhorn’s anchoring is the model; Lange’s point conceded by one parameter; commutativity given up on principle, not only through the reading.                                                                                                 |

What the comparison says about scope. The paper’s contribution lives in the regime where the sequence effect is subtle, arising in the rule rather than in the inputs, existing only because the traits are believed related, and therefore small and spread unevenly across statistics. That is the regime in which an audit can be fooled by choosing the wrong statistic. In the primacy regime the effect is blunt and any marginal shows it; the question becomes not whether the sequence registers but how much of the earlier impression survived, which is a question about  $\omega$ , answered by ADJ and nothing else. The two regimes are two different questions. Under full adoption the association audit fails by silence; under partial adoption it fails by fabrication as well.

## 6. Two paragraphs for Section 6, and the scope sentences

These expand the prelude’s last clauses, that the paper explores the full-adoption setting and that the adoption weight decides which channels operate. They follow 6.A in the plan (6.B), with Table ?? beside them. Every quantity is a row of Table ?? or a proposition already in the manuscript; the decision row of Table ??, which is inferred, is not used. Rubric scoring is a new element, an extension of the hiring-panel example, and no empirical claim is made about whether rubrics work.

**The two channels.** Sequence dependence has two channels in this setting, and the results of Sections 4 and 5 concern one of them. The first is the association channel. A cue on one attribute moves the belief about the other through the prior association, so the second cue meets a belief the first has already changed. This effect is first order in  $c$  and vanishes at independence, where each cue sets its own marginal and touches nothing else (Proposition IMM). The second is the position channel. If the cue read second is adopted only in part, then which cue is weakened depends on the order, and the two sequences differ even at  $c = 0$ , by  $(1 - \omega)(\alpha - q_0)$  on the competence marginal and  $-(1 - \omega)(\beta - r_0)$

on the trustworthiness marginal. The position channel is not a feature of updating on delivered credences. A Bayes-factor update that gives the second cue's factor the weight  $\omega$  is order-dependent in the same way, since commutation requires the same factor in either position (Wagner 2002). Under full adoption the position channel is absent and the association channel is the only one, which is the setting of the results. Under partial adoption the position channel is present at  $c = 0$  and moves every marginal at zeroth order, while the believed association still does not differ between sequences at  $c = 0$  and differs at first order; the classification by order in  $c$  shifts by one order without changing the ranking of statistics (Proposition LAD, corrected 2026-10-01), and the mean belief of a population mixing the two sequences acquires a cross-product association of  $-\lambda(1 - \lambda)(1 - \omega)^2(\alpha - q_0)(\beta - r_0)$  that no member holds. What the paper says about the position channel is confined to Propositions ADJ and LAD, the first recovering its strength from the ratings and the second stating what it does to the ranking. The position channel is a temporal bias of the observer, present whether or not the attributes are believed related, while the association channel runs through what one cue implies about the other and exists only when they are.

**Opposite predictions about one procedure.** Whether a procedure removes the sequence effect depends on which channel it removes. Consider rubric scoring, in which the rating on each attribute is fixed by the document on that attribute and a set scale, so that what each document delivers no longer depends on what was read first. Under full adoption the rubric removes the position channel and leaves the association channel untouched. The recorded scores carry the delivered credences in either sequence, the belief about the attribute read second agrees with its score, and the belief about the attribute read first sits at  $cK$  from its score (Proposition DRF), so the sequence effect lives in that belief and in the decision made from it, vanishes when the attributes are believed unrelated (Proposition IMM), and is invisible in the scores and in the association between them. Under partial adoption the rubric removes the position channel from the record only. The belief about the attribute read second sits at  $(1 - \omega)[P^A(B=1) - r_1]$  from its score (Proposition ADJ), which at independence equals  $(1 - \omega)(r_0 - \beta)$ , so the two reading sequences differ even when the attributes are believed unrelated. The two settings therefore predict opposite things about the same procedure, and the difference is observable. In the rubric setting  $r_1$  is on the record, so Proposition ADJ needs only the belief about trustworthiness after the credential alone and after both documents to say which prediction holds.

**Two sentences for the scope statement (approved 2026-09-27).** The position channel is a temporal bias of the observer, present whether or not the attributes are believed related, while the association channel runs through what one cue implies about the other and exists only when they are. Full adoption is the setting in which the only source of sequence dependence is the association channel, which vanishes at  $c = 0$ ; partial adoption adds the position channel, the weakening of whichever cue is read second, which is present at  $c = 0$  and so lies outside the expansion in  $c$  on which the results rest.

## 7. Notes for applying

**Duplication to watch.** Once everything is applied,  $(1 - \omega)(\alpha - q_0)$  would appear in ADJ, in Table ??, and in the two-channel paragraph. ADJ is the proposition and carries it; the table and the paragraph cite ADJ. 6.A already cites ADJ and does not repeat the expression. The pooled association distortion appears in Table ?? and the two-channel paragraph, both citing the verification, and nowhere else.

**What must not go in.** The decision row of Table ?? (share of flips and loss at interior  $\omega$ ) is inferred from linearity and not checked; it stays out of the manuscript. Nothing at interior  $\omega$  enters as a proposition; the interior numbers appear inside Section 6 prose and Table ??, as scope justification, each traceable to a row of the interior script.

**Order of application.** 5.B before Table ??, the two Section 6 paragraphs and the prelude, since all cite ADJ; B.A before any of them, for the bibliography.